Light

GCSE Computer Science — Memory and Storage (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A user opens 20 browser tabs simultaneously and notices their computer slowing down.
Which component is most likely running low?

- **A.** RAM
- **B.** ROM
- **C.** The monitor's refresh rate
- **D.** The keyboard buffer

Q2 (1 mark)

Why is ROM described as 'Read-Only Memory'?

- ○ **A.** Because it can only display text, not images
- ○ **B.** Because under normal operation, data can be read from it but not written to it
- ○ **C.** Because it can only be accessed by the operating system
- ○ **D.** Because it stores only one file at a time

Q3 (1 mark)

On a Windows computer, what is the file used for virtual memory typically called?

- ○ **A.** boot.ini
- ○ **B.** system32.dll
- ○ **C.** pagefile.sys
- ○ **D.** desktop.ini

Q4 (1 mark)

An HDD is advertised as '7200 RPM'.
What does this specification indicate?

- ○ **A.** The platters rotate at 7200 revolutions per minute, affecting data transfer speed
- ○ **B.** The drive can store 7200 megabytes
- ○ **C.** The drive has 7200 read/write heads
- ○ **D.** The drive costs £72.00

Q5 (1 mark)

Why is defragmenting an SSD unnecessary and potentially harmful?

- ○ **A.** SSDs automatically defragment themselves every hour
- ○ **B.** SSDs have no moving parts, so fragmented data can be accessed equally fast from any location, and extra writes reduce SSD lifespan
- ○ **C.** SSDs cannot store fragmented files
- ○ **D.** Defragmentation only works on optical drives

Q6 (1 mark)

Why have optical discs traditionally been popular for distributing commercial software and movies?

- ○ **A.** Optical discs have faster access than SSDs
- ○ **B.** Optical discs can store more data than hard drives
- ○ **C.** Optical discs are cheap to mass-produce, difficult for users to modify, and have standardised players worldwide
- ○ **D.** Optical discs never degrade over time

Q7 (1 mark)

How many bits are in a nibble?

- ○ **A.** 4 bits
- ○ **B.** 2 bits

- ○ **C.** 8 bits
- ○ **D.** 16 bits

Q8 (1 mark)

How many gigabytes (GB) are in 4 terabytes (TB)?

- ○ **A.** 4 GB
- ○ **B.** 400 GB
- ○ **C.** 40000 GB
- ○ **D.** 4000 GB

Q9 (1 mark)

A typical modern laptop has RAM measured in which unit?

- ○ **A.** Megabytes (MB)
- ○ **B.** Gigabytes (GB)
- ○ **C.** Terabytes (TB)
- ○ **D.** Kilobytes (KB)

Q10 (1 mark)

For archiving important data that must last 50+ years, which storage medium is often recommended?

- ○ **A.** USB flash drive
- ○ **B.** Standard hard disk drive
- ○ **C.** M-DISC (archival optical disc) or magnetic tape
- ○ **D.** RAM module

Q11 (1 mark)

An image has a resolution of 1920 × 1080.
What do these numbers represent?

- ○ **A.** The width and height in pixels
- ○ **B.** The file size in kilobytes
- ○ **C.** The colour depth in bits
- ○ **D.** The compression ratio

Q12 (1 mark)

In digital audio, what is a 'sample'?

- ○ **A.** A complete song or audio track
- ○ **B.** A single measurement of the sound wave's amplitude at a specific point in time
- ○ **C.** The file format of the audio
- ○ **D.** The volume setting of the playback device

Q13 (2 marks)

When a computer is first switched on, it needs instructions to start. Explain why these initial instructions must be stored in ROM rather than RAM.

- ○ **A.** ROM is faster than RAM
- ○ **B.** RAM has larger capacity than ROM
- ○ **C.** ROM can be modified more easily than RAM
- ○ **D.** RAM is empty when power is first applied because it's volatile; ROM retains its contents without power, so boot instructions are available immediately

Q14 (2 marks)

A computer has 8 GB of RAM and a virtual memory (page file) configured to 16 GB.
What is the maximum amount of memory that programs can theoretically use?

- ○ **A.** 8 GB (only physical RAM counts)
- ○ **B.** 16 GB (only virtual memory counts)
- ○ **C.** 24 GB (RAM plus virtual memory combined)
- ○ **D.** 128 GB (RAM multiplied by virtual memory)

Q15 (2 marks)

A file is 240 MB. An internet connection downloads at 8 MB per second. How long will the download take?

- ○ **A.** 30 seconds

- ○ **B.** 3 seconds
- ○ **C.** 300 seconds
- ○ **D.** 1920 seconds

Q16 (2 marks)

When an HDD needs to read data from a different part of the disc, there is a delay called 'seek time'.
What causes this delay?

- ○ **A.** The data must be decrypted before reading
- ○ **B.** The read/write head must physically move to the correct track, which takes time
- ○ **C.** The data must be downloaded from the internet
- ○ **D.** The operating system must grant permission to access the file

Q17 (2 marks)

Modern SSDs use a feature called 'TRIM'.
What does TRIM do?

- ○ **A.** Compresses files to save space
- ○ **B.** Removes the first and last sectors of each file
- ○ **C.** Reduces the SSD's capacity to extend its lifespan
- ○ **D.** Tells the SSD which data blocks are no longer in use, allowing the drive to optimise performance and manage wear levelling

Q18 (2 marks)

Blu-ray discs can store much more data than DVDs despite being the same physical size.
What enables this increased capacity?

- ○ **A.** Blu-ray uses a blue-violet laser with a shorter wavelength, allowing smaller pits and tighter tracks than DVD's red laser
- ○ **B.** Blu-ray discs spin ten times faster than DVDs
- ○ **C.** Blu-ray uses magnetic storage instead of optical
- ○ **D.** Blu-ray discs are made from a different type of plastic

Q19 (2 marks)

A character set needs to represent 500 different characters.
What is the minimum number of bits required per character?

- ○ **A.** 7 bits (128 values)
- ○ **B.** 8 bits (256 values)
- ○ **C.** 9 bits (512 values)
- ○ **D.** 10 bits (1024 values)

Q20 (2 marks)

An image's resolution is doubled in both width and height.
By what factor does the file size increase (assuming the same colour depth)?

- ○ **A.** The file size doubles (×2)

- B. The file size triples (×3)
- C. The file size increases by 8 times (×8)
- D. The file size quadruples (×4)

Q21 (2 marks)

CD audio uses a sample rate of 44,100 Hz.
Why was this specific value chosen?

- A. It was randomly selected as an easy number to remember
- B. It's slightly more than twice the upper limit of human hearing (~20,000 Hz), satisfying the Nyquist theorem for accurate sound reproduction
- C. It matches the refresh rate of old television screens
- D. It's the maximum rate that early CD players could handle

Q22 (2 marks)

External portable SSDs are becoming popular alternatives to USB flash drives.
What is the main advantage of portable SSDs over USB flash drives?

- A. Portable SSDs are physically smaller
- B. Portable SSDs are cheaper per gigabyte
- C. Portable SSDs offer larger capacities and faster transfer speeds
- D. Portable SSDs work without a USB connection

Q23 (2 marks)

A program uses an 8-bit unsigned variable to count items. The count reaches 255 and the program adds 1 more item.
What value will the variable display?

- ○ **A.** 256
- ○ **B.** 255 (it stops increasing)
- ○ **C.** -1
- ○ **D.** 0 (it wraps around due to overflow)

Q24 (3 marks)

An uncompressed video frame is 1920 × 1080 pixels with 24-bit colour. The video plays at 30 frames per second.
Calculate the data rate (amount of data per second) in MB/s.

- ○ **A.** Approximately 186.6 MB/s
- ○ **B.** Approximately 62.2 MB/s
- ○ **C.** Approximately 1.9 MB/s
- ○ **D.** Approximately 5.9 MB/s

Q25 (3 marks)

Some computers use 'hybrid drives' (SSHD) that combine an HDD with a small amount of flash memory.
Analyse how this design attempts to offer benefits of both technologies.

- ○ **A.** The flash memory stores the operating system permanently while the HDD stores user files

○ **B.** The HDD and flash memory store identical copies of all data for redundancy

○ **C.** The flash memory acts as a cache for frequently accessed data, providing SSD-like speed for common operations while the HDD provides large capacity at lower cost

○ **D.** The flash memory increases the HDD's spinning speed

Q26 (3 marks)

A system uses DDR4 RAM, which transfers data on both the rising and falling edges of the clock signal.
Analyse how this 'double data rate' design improves performance compared to earlier single data rate (SDR) RAM.

○ **A.** DDR4 uses half the clock speed of SDR RAM for the same performance

○ **B.** DDR4 transfers twice as much data per clock cycle as SDR because it uses both edges of the clock signal, effectively doubling bandwidth without increasing clock frequency

○ **C.** DDR4 stores twice as many bits in each memory cell

○ **D.** DDR4 reduces latency by half compared to SDR

Q27 (3 marks)

A music file can be stored in either WAV format (uncompressed) or FLAC format (lossless compression).
Analyse the differences between these formats.

○ **A.** WAV files sound better because they are uncompressed

○ **B.** FLAC permanently removes audio data that humans cannot hear, reducing file size

○ **C.** FLAC files are larger than WAV files due to added metadata

○ **D.** Both formats preserve the original audio data exactly, but FLAC uses compression algorithms to reduce file size by 50-60% without any loss of quality — when decoded, the audio is bit-for-bit identical to the original

Q28 (3 marks)

A system administrator is concerned that using an SSD for virtual memory might wear out the drive faster than using an HDD.
Analyse whether this concern is valid and what factors should be considered.

○ **A.** The concern is valid because virtual memory causes constant write operations, and SSDs have limited write cycles; however, modern SSDs have sufficient endurance for typical use, and the significant performance benefit usually outweighs the modest additional wear

○ **B.** The concern is completely invalid — SSDs never wear out

○ **C.** SSDs should never be used for virtual memory under any circumstances

○ **D.** HDDs wear out faster than SSDs when used for virtual memory

Q29 (3 marks)

A photograph is saved as a JPEG with high compression. The image shows visible 'artefacts' — blocky patterns and colour banding that weren't in the original.
Analyse what causes these artefacts and why they appear.

○ **A.** The camera sensor was damaged when taking the photograph

○ **B.** The monitor cannot display all the colours in the image

○ **C.** JPEG uses lossy compression that permanently discards image data; high compression aggressively removes detail to reduce file size, and the mathematical approximations used create visible errors when there isn't enough data to accurately reconstruct the original image

○ **D.** The file became corrupted during saving

Q30 (3 marks)

Analyse why solid-state storage has largely replaced magnetic hard drives in laptops and smartphones, despite HDDs offering larger capacities at lower prices.

○ **A.** SSDs are always cheaper than HDDs regardless of capacity

○ **B.** HDDs cannot physically fit inside laptops or smartphones

○ **C.** Consumers don't need storage capacity in mobile devices

○ **D.** For mobile devices, SSD advantages in durability (no moving parts, shock resistance), power consumption (lower battery drain), size/weight (no platters/motors), and speed (faster boot and app loading) outweigh HDD's cost-per-gigabyte advantage, making SSDs essential for modern portable device requirements

Total: **55** marks • Time: **30 min**

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